

B.Tech. Degree VI Semester Examination April 2016

CS 1603 OPERATING SYSTEM (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Define operating system. List out any five services provided by operating system.
 (b) What are semaphores? What are the operations associated with semaphore?
 (c) Distinguish between multiprogramming with fixed partitions and multiprogramming with variable partitions.
 (d) What is meant by memory compaction?
 (e) Define file attribute. List any five file attributes.
 (f) What are the features of RAID?
 (g) Explain the conditions for deadlock.
 (h) Explain Banker's algorithm.

PART B

(4 × 15 = 60)

- II. (a) What are processes? Explain the different process states, with a diagram. (5)
 (b) Explain race condition with an example. (10)
OR
 III. Write about process scheduling algorithms with example. (15)
 IV. (a) Compare the bitmaps and linked list memory management schemes. (8)
 (b) Describe swapping. (7)
OR
 V. Under what circumstances do page faults occur? With suitable diagram describe the actions taken by the operating system when a page fault occurs. (15)
 VI. (a) Explain the different file system implementation mechanisms. (10)
 (b) Explain the function of device drivers. (5)
OR
 VII. List and describe the RAID levels. (15)
 VIII. (a) Differentiate deadlock and starvation. (5)
 (b) Describe the methods of deadlock detection and recovery. (10)
OR
 IX. Consider the following snapshot of a system. (15)

	Allocation	Max	Available
	A B C D	A B C D	A B C D
P0	0 0 1 2	0 0 1 2	1 5 2 0
P1	1 0 0 0	1 7 5 0	
P2	1 3 5 4	2 3 5 6	
P3	0 6 3 2	0 6 5 2	
P4	0 0 1 4	0 6 5 6	

Answer the following questions using banker's algorithm.

- (i) What is the content of the matrix need?
 (ii) Is the system in safe state?
 (iii) If a request from process P1 arrives for (0, 4, 2, 0), can the request be granted immediately?

